FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2015

ANSWERS, HINTS & SOLUTIONS
FULL TEST–II
(PAPER-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	B	D
2.	B	C	B
3.	D	D	D
4.	A	A	A
5.	A	A	A
6.	A	A	D
7.	C	C	B
8.	B, C, D	B, C, D	A, B, C, D
9.	A, B, C, D	A, C	A, C
10.	A, C, D	A, B, C, D	A, B
11.	A, B, C	A, D	A, B, C, D
12.	A	B	C
13.	B	B	C
14.	A	B	D
15.	C	C	C
16.	D	C	B
1.	(A)→ (r), (B)→ (q), (C)→ (p), (D)→ (p)	(A → q) (B → s) (C → r) (D → p)	(A) → (s) (B) → (p) (C) → (r) (D) → (t)
1.	(A) → (p, s, t), (B) → (q, r), (C) → (q, r), (D) → (p, s, t)	(A) → (q), (B) → (s), (C) → (r), (D) → (p)	(A) → (p) (B) → (r) (C) → (p) (D) → (r)
1.	5	4	9
2.	2	7	1
3.	2	n = 3	5
4.	5	9	8
5.	2	8	1
6.	4	3	2

Physics

PART – I

SECTION – A

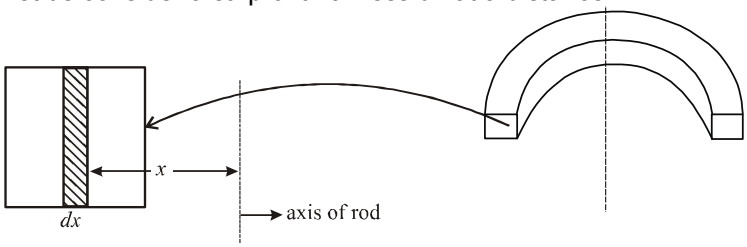
1. Electric field is tangential direction at all points $(\vec{E}.\hat{r}=0)$ and decreasing with increase in r. (r = magnitude of radius vector of a point)

2. Since, Electrostatic potential is zero at all points on axis,
So, along axis $\frac{\partial v}{\partial r}=0 \Rightarrow E_r = 0$

3. $V_a = V_b = V_c$ because electric field inside conductor is zero.
$V_b = \int_{\infty}^{\vec{r}_b}(\vec{E}+\vec{E}'+\vec{E}_{ind}+\vec{E}_{ind}).\vec{dr} = \int_{-\infty}^{\vec{r}_b}(\vec{E}'+\vec{E}'_{ind}).\vec{dr}$, because $\vec{E}'+\vec{E}'_{ind}=0$ for all points inside conductor or outside the conductor.

4. An infinite wire can be taken as half of closed loop.
Hence $\int_{-\infty}^{\infty}\vec{B}\cdot\vec{d\ell}=\frac{\mu_0 i}{2}$

5. Let us consider a strip of thickness dx at a distance 'x'



$dR=\frac{\rho.\pi x}{a dx}$

$\frac{1}{R_{eff}}=\int\frac{1}{dR}=\frac{a}{\pi\rho}\int_r^{r+a}\frac{dx}{x} \Rightarrow R_{eff}=\frac{\pi\rho}{a\ln\left(1+\frac{a}{r}\right)}$

6. $V_A + V_{ind} = 0 \Rightarrow K.\frac{q}{(d-a\sin\omega t)}+\frac{K q_{ind}}{R}=0$

$\Rightarrow q_{ind} = -\frac{R.q}{(d-a\sin\omega t)} \Rightarrow \frac{dq_{ind}}{dt}=\frac{Rqa\omega\cos\omega t}{(d-a\sin\omega t)^2}$

8. Force applied by hinge on rod may be towards P, away from P or zero depending upon distance from point 'O'. Accordingly linear momentum of rod + bullet system may decrease increase or remains constant respectively. Similarly angular momentum of system about 'Q' may decrease, increase or remain constant respectively.

9. λ = 1 m
$v_{max} = A\omega = 4\pi$
$\therefore \quad v=\frac{\partial y/\partial t}{\partial y/\partial x}=\frac{4\pi\times10^{-2}}{4\pi\times10^{-3}}=10\,m/s$

A = 2 × 10^{-3}m also $\nu=\frac{v}{\lambda}=10\,Hz$ and $a_{max}=A\omega^2=80\pi\,cm/s^2$

10. From centre of mass of system ball and centre of tube moves in circular path.

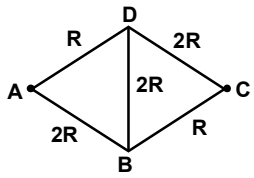
Radius of curvature of ball at the time of projection of ball is $\frac{MR}{m+M}$

Normal force between tube and ball if M = 2m, at the time of projection of ball is

$V_{cm} = \frac{mV_0}{m+M}$

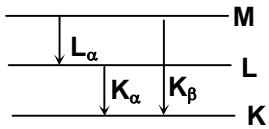
$N = m\left(\frac{(V_0 - V_{cm})^2}{r_1}\right)$

11. Equivalent circuit is



14. $\frac{1}{\lambda_{k\beta}} = \frac{1}{\lambda_{k\alpha}} + \frac{1}{\lambda_{L\alpha}}$

$\Rightarrow \lambda_{L\alpha} = \frac{\lambda_{k\alpha} \cdot \lambda_{k\beta}}{\lambda_{k\alpha} - \lambda_{k\beta}} = 5.60\text{Å}$



15. $eV = \frac{hc}{\lambda_{cut\,off}} \Rightarrow V = \frac{hc}{e\lambda_{cut\,off}} = \frac{6.626\times10^{-34}\times3\times10^{8}}{1.6\times10^{-19}\times0.4\times10^{-10}} = 31\times10^{3}\ V$

SECTION - B

1. Using Kirchof's first law at point P

SECTION – C

1. $V_O = V_P$

$\frac{KQ}{R} + \frac{KQ}{3R} = V_{shell} + \frac{2Kq}{5R}$

$V_{shell} = \frac{KQ}{R}\left[1 + \frac{1}{3} - \frac{2}{5}\right]$

$= \frac{KQ}{R}\left[\frac{4}{3} - \frac{2}{5}\right]$

$= \frac{KQ}{R}\left[\frac{20-9}{15}\right] = \frac{11}{15}\frac{KQ}{R}$

2. Using Bernoulli's theorem

$\frac{1}{2}\rho v^2 = p = \rho gh \quad \text{or} \quad h = \frac{v^2}{2gh}$ ..... (1)

Put : $V = r\omega = r(2\pi\nu)$ in (1)

5. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v}-\frac{1}{(3R/2)}=-\frac{2}{R}$$

$$\frac{1}{v}=-\frac{2}{R}+\frac{2}{3R}=-\frac{4}{3R}$$

$$v=-\frac{3R}{4}$$

$$m=-\frac{v}{u}=-\frac{\left(-\frac{3R}{4}\right)}{\left(-\frac{3R}{2}\right)}=-\frac{2}{4}=-\frac{1}{2}$$

amplitude of image = $4\left(-\frac{1}{2}\right)=2$

Chemistry

PART – II

SECTION – A

1. $Ag\,|\,Ag_2CO_3(s)\,|\,Na_2CO_3(aq.)\,||\,KBr(aq.)\,|\,AgBr(s)\,|\,Ag(s)$

Anode : $2Ag(s) + CO_3^{2-} \longrightarrow Ag_2CO_3(s) + 2e^-$

Cathode : $AgBr(s) + e^- \longrightarrow Ag(s) + Br^-(aq.)$

$CO_3^{2-}(aq.) + 2AgBr(s) \longrightarrow 2Br^-(aq.) + Ag_2CO_3(s)$

$$E_{cell} = E^o_{cell} - \frac{0.059}{2}\log\frac{\left[Br^-\right]^2}{\left[CO_3^{2-}\right]}$$

At eqm.

$$0 = E^o_{cell} - \frac{0.059}{2}\log\frac{\left[Br^-\right]^2}{\left[CO_3^{2-}\right]}$$

$$0 = E^o_{cell} - 0.059\ \log\frac{\left[Br^-\right]}{\sqrt{\left[CO_3\right]^{2-}}}$$

$$0.059\log\frac{\left[Br^-\right]}{\sqrt{\left[CO_3^{2-}\right]}} = E^o_C - E^o_A$$

$$E^o_{Ag/AgCl/Cl^-} = E^o_{Ag^+/Ag} + 0.059\ \log K_{spAgBr}$$

$$E^o_{Ag/Ag_2CO_3/CO_3^{2-}} = E^o_{Ag^+} + 0.059\ \log\frac{K_{sp}\ Ag_2CO_3}{K_{sp}\ AgBr}$$

$$\log\frac{\left[Br^-\right]}{\sqrt{CO_3^{2-}}} = \log\frac{K_{sp}\ AgBr}{K_{sp}\ Ag_2CO_3}$$

$$\frac{Br^-}{\sqrt{\left[CO_3^{2-}\right]}} = \frac{4\times10^{-13}}{\sqrt{8\times10^{-12}}} = \sqrt{2}\times10^{-7}$$

2. $\lambda_1 N + \lambda_2 N = \lambda N$

$$\lambda_1 = \frac{\ln 2}{24} \qquad \lambda_2 = \frac{\ln 2}{6}$$

$\lambda N_o = 6$

$\lambda N_o e^{-\lambda t} = 3$

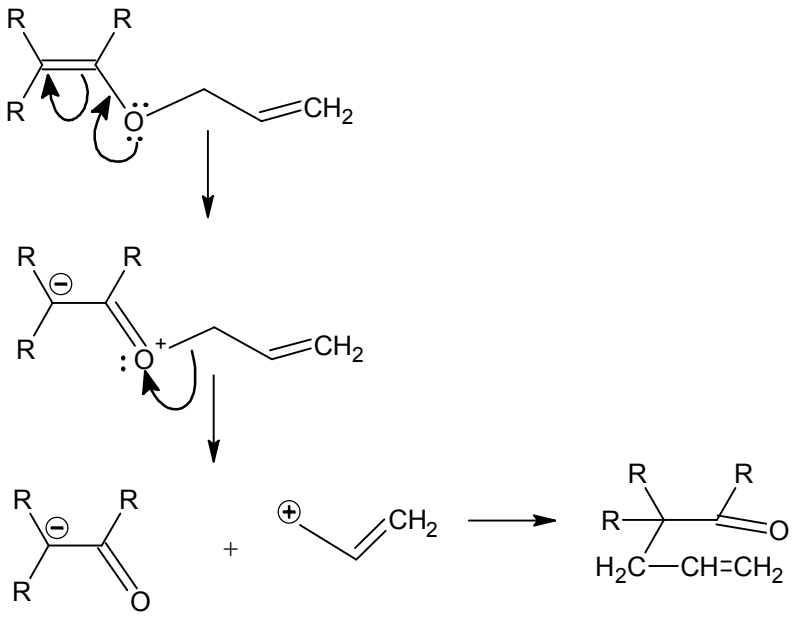
$e^{\lambda t} = 2$

$\lambda t = \ln 2$

$$\left(\frac{\ln 2}{24} + \frac{\ln 2}{6}\right)t = \ln 2$$

t = 4.8 hours

3.



4. $\frac{r^+}{r^-} = 0.414$ for octahedral void fitted with our red spherical balloon.

$\frac{r^+}{\alpha} = 0.414$

$r^+ = 0.414\alpha$

Initial r^+ radius = 0.6α

Final r^+ radius = 0.414α

$\frac{\text{Initial r}}{\text{Final r}} = \frac{6\alpha}{0.414\alpha} = 14.5$

5. The number of maxima in the radial charge density curve is (n – *l*)
For 3d electron, n = 3 and d = 2
Hence, n – *l* = 3 – 2 = 1
There will be only one maximum in the curve for 3 d electron.

6. In Perkin reaction, the carbanion attacks the carbonyl carbon of aromatic aldehyde to produce finally the unsaturated acid
The electron withdrawing groups increase reactivity of aldehydes in this reaction by increasing the electron deficiency of carbonyl atom. Thus $-NO_2$ increases the reactivity and the groups like $-N(CH_3)_2$ and $-CH_3$ decreases the reactivity.

7. The order of ligand strength in the spectrochemical series $F^- < H_2O < NH_3 < NO_2^-$. A strong ligand causes a larger degree of splitting resulting in high value of E (energy). Therefore corresponding low value of $\lambda \left[E = \frac{hc}{\lambda} \right]$

Protonation occurs on the OH which can produce more stable carbocation and while rearrangement phenyl has higher migratory aptitude over o-substituted phenyl.

Solutions for Q. No. 12 to 13

Let the no. of moles of H_2, O_2 and He are x, y and z respectively

$\therefore x + y + z = 2.0$

Pressure exerted by 2 moles of gaseous mixture is 50 atm.

After the first electric spark decrease in pressure = 50 – 12.5 = 37.5 atm

$\therefore$ decrease in no. of moles of gaseous mixture is 1.5

But from the given information limiting reagent in first step is O_2

$\therefore$ from $H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$

$\therefore$ H_2 and O_2 reacted are in the ratio of 2 : 1

$\therefore$ 0.5 mole of O_2 should be reacted with 1.0 mole of H_2

$\therefore$ No. of mole of O_2 in the mixture (y) = 0.5

When again O_2 is passed, pressure increased from 12.5 to 25.0 atm

$\therefore$ Change in pressure = 12.5 atm

$\therefore$ No. of moles of O_2 added = $\frac{12.5 \times 2}{50} = 0.5$ mole

Now H_2 will be completely reacted

No. of moles H_2 actually left after first electric spark = x – 1

$$H_2 \quad + \quad \frac{1}{2}O_2 \quad \longrightarrow H_2O$$

$$(x-1) \quad \left(\frac{x-1}{2}\right)$$

Now change in pressure in 25 – 10 = 15 atm

$\therefore$ change in no. of moles is $\frac{15 \times 2}{50} = 0.6$

$\therefore (x-1) + \left(\frac{x-1}{2}\right) = 0.6$

$\Rightarrow x = 1.4$

$\therefore$ no. of moles of H_2 = 1.4

$\therefore$ no. of moles of He = 2.0 – (1.4 + 0.5) = 0.1

$\therefore$ mole fraction of H_2 = $\frac{1.4}{2.0}$ = 0.7

14. Possible structure after test (a) would be that of a 1° or a 2° alcohol and not that of 3° alcohol.

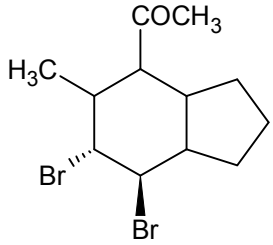
15. Only non-resolvable compound is a 1° alcohol out of the given choices.

16. Oxidation of compound with CrO_3 in pyridine gives a compound, that gives Tollen's test. This implies that it is a primary alcohol.

SECTION-B

1. Silver→ Ore is dissolved by complex formation using $\bar{C}N$ in presence of air followed by precipitation with zinc.
   Lead → Involves roasting of PbS followed by self reduction.
   Iron → Involves calcination followed by carbonmonoxide reduction.
   Magnesium → Involves calcination (of $MgCl_2.6H_2O$ in presence of dry HCl gas) followed by fused salt electrolysis.

SECTION –C

2. P = 

3. $n_{M^{n+}} = \frac{1.92}{2\times 60}$ mol = 0.016 mol.

$n_{hydrazinehydrate} = = \frac{0.9\times 5.332}{50}$ mol = 0.096 mol.

$n_{eqKIO3} = 4\times\frac{1}{10}N\times 0.48L = 0.192 \simeq n_{eqhydrazine}$ reacted with KIO_3

$\therefore n_{hydrazine}$ left after reaction with $M^{n+} = \frac{0.192}{4}$ mol = 0.048 mol

$\therefore n_{hydrazine}$ reacted with M^{n+} = (0.096 – 0.048) mol = 0.048 mol

$\frac{n_{M^+}}{n_{hydrazine}} = \frac{1}{n} = \frac{0.016}{0.048} = \frac{1}{3}$

4. F T T T T F F F T T F T T T
(IX) A is a buffer solution; pH of A = pK_a = 2

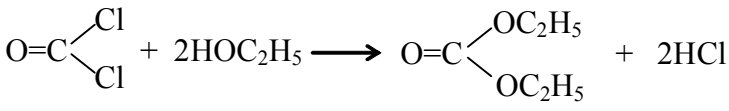
B is an aqueous solution of SacNa; pH of B = $\frac{1}{2}(pK_w + pK_a + \log c)$ = 8

5. Total work done = Area of circle A (anticlockwise) – Area of circle B (clockwise)
= 40 – 36 = 4 unit
heat flows into the system along the path ADB, = 5 – 3+2 = 4 unit

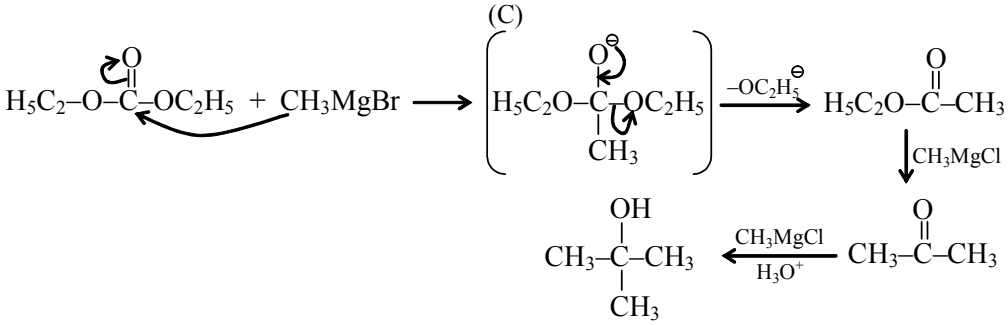
6. $CCl_3–CH(OH)_2$ + NaOH $\longrightarrow$ $CHCl_3$ + $HCOO^-Na^+$ + H_2O
Chloral hydrate (A)

$CHCl_3 + \frac{1}{2}O_2 \xrightarrow{h\nu} COCl_2 + HCl$

(A) (phosgene)
(B)



(diethyl carbonate)
(C)



Mathematics

PART – III

SECTION – A

1. Possibilities:

   (i) three digit occur twice = $^5C_3 \cdot \frac{6!}{2!2!2!} = 900$ ways

   (ii) two digit occur three times = 200 ways

   (iii) one digit occur four times and other twice = 300 ways

   (iv) all digits are same = 5

   $\Rightarrow$ Total possibilities = 1405

2. $|z - 2i| = 2$ represent a circle with centre (0, 2), radius 2 unit and $|z - 1 - 3i| + |z + 1 - 3i| = 2\sqrt{2}$ represent an ellipse they touch each other at (0, 4)

3. Let $f(n) = \sum_{r=0}^{n} {}^{n+1}C_{r+1} = 2^{n+1} - 1$

   f(9) = 1023 = 3.11.31

   Total divisors are = 8

4. $2\left(1+x^3\right)^{100} = \left[a_0 + a_1x + a_2x^2 + ... + a_{100}x^{100}\right] - \left[\cos\frac{\pi}{2}x + \cos\left(\frac{\pi}{2}x + \frac{\pi}{2}\right) + ... + \cos\left(\frac{\pi}{2}x + 100\frac{\pi}{2}\right)\right]$

   Replacing x by –x and adding

   $\left(1+x^3\right)^{100} + \left(1-x^3\right)^{100} = \left[a_0 + a_2x^2 + ... + a_{100}x^{100}\right] - \left(\cos\frac{50\pi}{2}\cos\frac{\pi x}{2}\right)$

   Put x = 1 $\Rightarrow$ $2^{100} = \sum_{i=0}^{50} a_{2i}$ $\Rightarrow$ k = 100

5. The desired condition is to have the circle N inside the convex region of the parabola $y = x^2$ clearly $a \geq 1$, otherwise circle will cross x-axis also for circle to be inside the parabola the number of points of intersection must be less than or equal to 2

   $\Rightarrow$ Equation $y + (y - a)^2 = 1$, has one or less than one solution

   $\Rightarrow D \leq 0$

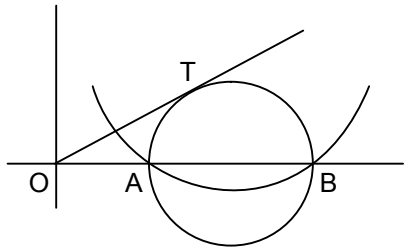
   $\Rightarrow a \geq \frac{5}{4}$

6. $I_N = \int_0^{\pi} x^3 \log_e(\sin x)\,dx = \int_0^{\pi} (\pi - x)^3 \log_e(\sin x)\,dx$

   $= \frac{3\pi}{2}\int_0^{\pi} x^2 \log_e\sqrt{2}\,dx + \frac{3\pi}{2}\int_0^{\pi} x^2 \log_e(\sin x)\,dx = \frac{3\pi}{2}I_D$

   $\Rightarrow \frac{I_N}{I_D} = \frac{3\pi}{2}$

7. $OT^2 = OA \cdot OB$



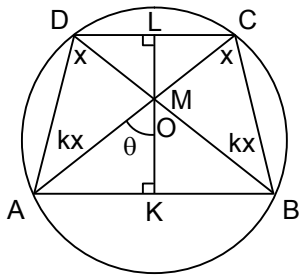
8. Let ABCDPQRS be a cube, the numbers a, b, c, d, e, f be written on the faces
$\Rightarrow$ ade + abe + abf + adf + cde + bce + bcf + cdf = 2004
$\Rightarrow$ (e + f)(a + c)(b + d) = $2^2 \cdot 3 \cdot 167$
Possible combination are $4 \cdot 3 \cdot 167$, $2 \cdot 6 \cdot 167$, $2 \cdot 3 \cdot 334$, $2 \cdot 2 \cdot 501$
T = a + b + c + d + e + f = 174, 175, 339, or 505

9. If $x \geq 0$, $x^2 + (a - 5)x + 1 = 0$
$x < 0$, $x^2 + (a + 1)x + 1 = 0$
Equation has exactly three distinct solution if (1)
Either $(a - 5)^2 > 4$ and $(a + 1)^2 = 4$
Or $(a - 5)^2 = 4$ and $(a + 1)^2 > 4$

10. $\Delta ODC = \Delta HEA = \Delta GFB$
$$\Rightarrow \frac{1}{2}R^2 \sin A \cos A = 2R^2 \sin C \cos C \cos^2 A = \frac{1}{3}R^2 \sin A \sin B \sin C$$
First two equality gives, $\tan A = 4 \sin C \cos C$
From first and last, $3 \cos A = 2 \sin^2 C (1 + 4 \cos^2 C) \cos A$
$\Rightarrow 3 = 2 \sin^2 C (1 + 4 \cos^2 C)$
$$\Rightarrow \sin C = \frac{\sqrt{3}}{2} \text{ or } \frac{1}{\sqrt{2}}$$

11. Score of 5 can be obtained as 5 or (1, 4) or (1, 1, 3) or (1, 1, 1, 2)
P (scoring 5) = $\frac{1}{6} + \frac{1}{6^2} + \frac{1}{6^3} + \frac{1}{6^4}$ similarly
P (scoring 6) = $\frac{1}{6} + \frac{1}{6^2} + \frac{1}{6^3} + \frac{1}{6^4} + \frac{1}{6^5}$
P (scoring 7) = $\frac{1}{6^2} + \frac{1}{6^3} + \frac{1}{6^4} + \frac{1}{6^5} + \frac{1}{6^6}$
P (scoring 1) is impossible event

12.-13. Let OM = d
$\Delta MDC \sim \Delta AMB$
$AK^2 + OK^2 = AO^2 = DO^2 = DL^2 + OL^2$
$\Rightarrow k^2x^2 \sin^2\theta + (kx \cos\theta - d)^2 = x^2 \sin^2\theta + (x \cos\theta + d)^2$
$\Rightarrow (k^2 - 1)x^2 = 2xd (k + 1) \cos\theta$ as $k + 1 > 0$
$\Rightarrow (k - 1)x = 2d \cos\theta$
Now, $AB - CD = 2(AK - LD) = 2(k - 1)x \sin\theta = 2d \sin 2\theta$
If $\angle AMB = 60^\circ$, $\theta = 30^\circ$



If $\angle AMD = 60^\circ$, $\theta = 60^\circ$ in either case $\sin 2\theta = \frac{\sqrt{3}}{2}$
$AB - CD = 2\sqrt{3}$

14.-16. a, c, b are in H.P.
$$\Rightarrow \frac{2}{c} = \frac{1}{a} + \frac{1}{b}$$
Line $\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0$ passes through a fixed point $\left(-\frac{1}{2}, -\frac{1}{2}\right)$
So centre of ellipse is $\left(-\frac{1}{2}, -\frac{1}{2}\right)$
Which is also centre of auxiliary circle
$\Rightarrow p = \frac{1}{2}$, $q = \frac{1}{2}$ radius of auxiliary circle = 2

$\therefore$ Length of major axis = 4 also it segment of focal chord are 1, 2

$\Rightarrow \frac{1}{1}+\frac{1}{2}=\frac{4}{b^2} \Rightarrow b^2=\frac{8}{3}$

Director circle $\left(x+\frac{1}{2}\right)^2+\left(y+\frac{1}{2}\right)^2=\frac{8}{3}+4$

SECTION – B

1. (A) 20, b, c are in H.P.

$\Rightarrow \frac{1}{20}+\frac{1}{c}=\frac{2}{b}$

This can be written as (40 – b)(c + 20) = 800 we must have 20 < b < 40 or 0 < 40 – b < 20, consider factorisation of 800 in which one term is less that 20

(40 – b)(c + 20) = 800 = 1 $\times$ 800 = 2 $\times$ 400 = 4 $\times$ 200 = 5 $\times$ 160 = 8 $\times$ 100 = 10 $\times$ 80 = 16 $\times$ 50

Thus pair (b, c) = (39, 780), (38, 380), (36, 180), (35, 140), (32, 80), (30, 60), (24, 30) but b divides c

$\Rightarrow$ Only 5 triples satisfy required requirement

(B) Solutions of m + n + p = 10, are ${}^{3+10-1}C_{3-1}$

(C) ab + b – a + 1 = 0

$\Rightarrow$ (a + 1)(b – 1) = –2

$\Rightarrow$ (a + 1, b – 1) $\in$ {(1, –2), (–1, 2), (2, –1), (–2, 1)}

$\Rightarrow$ (a, b) $\in$ {(0, –1), (–2, 3), (1, 0), (–3, 2)

(D) Let z = x + iy then $(x + iy)^{20} = x – iy$

$\Rightarrow |z|^{20} = |z| \Rightarrow |z|(|z|^{19} – 1) = 0$

$\Rightarrow |z| = 1$ or 0

$\Rightarrow |z| = 0$

$\therefore$ Ordered pair (0, 0)

Let $z \neq 0$ then $(x + iy)^{20} (x + iy) = x^2 + y^2$

$\Rightarrow z^{21} = |z|^2 = 1 \Rightarrow z^{21} – 1 = 0$, which are 21 roots

2. (A) $f(x) = x^4 + ax^3 + bx^2 + cx + d$

$\alpha, \beta, \gamma, \delta$ are real roots of equation f(x) = 0

$\Rightarrow f(x) = (x – \alpha)(x – \beta)(x – \gamma)(x – \delta)$

$\Rightarrow |f(i)| = 1 \Rightarrow (1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)(1 + \delta^2) = 1$ this is possible if $\alpha = \beta = \gamma = \delta = 0$

$\Rightarrow a = b = c = d = 0$

(B) $x^5 – 1 = 0$

$\Rightarrow (x – 1)(x^4 + x^3 + x^2 + x + 1) = 0$

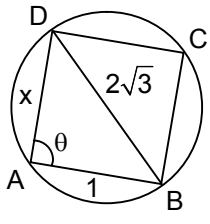
$$\Rightarrow \frac{x}{1+x^2}+\frac{x^2}{1+x^4}+\frac{x^3}{1+x}+\frac{x^4}{1+x^3}=\frac{x}{1+x^2}+\frac{x^2}{1+x^4}+\frac{x^3}{x^5+x}+\frac{x^4}{x^5+x^3}$$

$$= \frac{x}{1+x^2}+\frac{x^2}{1+x^4}+\frac{x^2}{1+x^4}+\frac{x}{1+x^2} = 2\left(\frac{x}{1+x^2}+\frac{x^2}{1+x^4}\right) = \frac{2\left(x+x^5+x^2+x^4\right)}{1+x^2+x^4+x^6}=2$$

(C) $\left[\cos|x|+\frac{1}{2}\right]>0 \Rightarrow \cos|x|+\frac{1}{2}\geq 1$ and $-\frac{1}{2}\leq \cos|x|+\frac{1}{2}\leq \frac{3}{2}$

$$\Rightarrow \left[\cos|x|+\frac{1}{2}\right]=1 \Rightarrow f(x) = 0 \Rightarrow \int_{x_1}^{x_2} f(x)=0$$

(D) Let AD = x, $\angle BAD = \theta$



In ΔABD, $\frac{BD}{\sin\theta} = 2R$

$\Rightarrow \sin\theta = \frac{2\sqrt{3}}{4} \Rightarrow \theta = 60^\circ$

Using cosine rule $\cos 60^\circ = \frac{x^2 + 1 - BD^2}{2x}$

$\Rightarrow BD = \frac{1+3\sqrt{5}}{2}$

$\therefore a = \frac{1}{2}, b = \frac{3}{2}$

SECTION – C

1. $f(9) = f(4 + 5) = f(4.5) = f(20)$
$= f(16 + 4) = f(16.4) = f(64)$
$= f(8.8) = f(8 + 8) = f(16)$
$= f(4.4) = f(4 + 4) = f(8)$

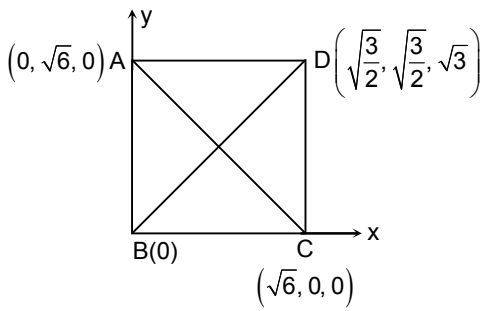
2. $g(x) = f'(x) - f(x)$ ..... (1)
$g'(x) = f'(x) - 2f(x)$ ..... (2)
$F'(x) = (2f(x)f'(x) + 2g(x)g'(x))(f^2(-x) + g^2(-x)) + (f^2(x) + g^2(x))(-2f(-x)f'(-x) - 2g(-x)g'(-x))$ ..... (3)
Using (1) and (2) in equation (3), we get $F'(x) = 0$
$\Rightarrow F(x)$ is constant function $\Rightarrow F(0) = 1 = F(1)$

3. $f(x) = \tan^{-1}(\tan x)$
$\sum_{r=1}^{5} f(r) = f(1) + f(2) + f(3) + f(4) + f(5) = 1 + 2 - \pi + 3 - \pi + 4 - \pi + 5 - 2\pi = 15 - 5\pi$
$a = 15, b = 5$

4. Minimum area will be in case of square with vertices
$(-\sqrt{2}, \sqrt{2})$, $(\sqrt{2}, \sqrt{2})$, $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$
$\therefore$ Minimum area = 8s

5. $\sum_{r=0}^{200} \alpha_r x^r = \sum_{r=0}^{200} \beta_r (1+x)^r$
$\alpha_0 + \alpha_1 x + ... + \alpha_{200} x^{200} = \beta_0 + \beta_1(1+x) + ... + \beta_{200}(1+x)^{200}$
$= \beta_0 + \beta_1(1+x) + ... + \beta_{99}(1+x)^{99} + \frac{(1+x)^{201} - (1+x)^{100}}{x}$
Equating coefficient of x^{100}, x^{101}, ..., x^{200} respectively we get
$\alpha_{100} = {}^{201}C_{101}$, $\alpha_{101} = {}^{201}C_{102}$, ..., $\alpha_{200} = {}^{201}C_{201}$
$\Rightarrow A = \sum_{r=100}^{200} \alpha_r = \alpha_{100} + \alpha_{101} + ... + \alpha_{200} = 2^{200}$
When 2^{200} is divided by 15 we get remainder 1

6. Take B as the origin, BC and BA as x and y axis and z axis perpendicular to the plane of ABC



$\Rightarrow \overrightarrow{BA} = \sqrt{6}\hat{j}$, $\overrightarrow{CD} = \left(\sqrt{\frac{3}{2}} - \sqrt{6}\right)\hat{i} + \sqrt{\frac{3}{2}}\hat{j} + \sqrt{3}\hat{k}$

$\Rightarrow \vec{n} = \overrightarrow{BA} \times \overrightarrow{CD} = 3\hat{k} + 3\sqrt{2}\hat{i} = 3\left(\sqrt{2}\hat{i} + \hat{k}\right)$

The minimum distance d = $\left|\overrightarrow{BC} \cdot \frac{\vec{n}}{|\vec{n}|}\right| = \frac{2}{\sqrt{3}} \cdot \sqrt{3} = 2$